Prac 4

1. Aim of the practical

To perform K-Means clustering on the Iris dataset and find the optimal number of clusters using the Elbow Method.

In simple words, we group similar flowers together based on their measurements.

2. What is clustering?

Clustering is an unsupervised machine learning technique used to divide data into groups based on similarity.

For example, imagine 150 flowers. We want to group flowers that have similar measurements together, without telling the algorithm their actual species names.

- Similar data points are placed in the same cluster.
- Different data points are placed in different clusters.

Remember: Clustering is unsupervised learning because it does not require labelled output to create groups.

3. What is the Iris dataset?

The Iris dataset contains information about three types of Iris flowers.

Your practical uses 150 flower records and four main features:

Feature	**Meaning**
SepalLengthCm	Length of sepal
SepalWidthCm	Width of sepal
PetalLengthCm	Length of petal
PetalWidthCm	Width of petal

The dataset has:

- 150 rows (flower records).
- 4 numerical features.
- 3 actual species.
- 50 records for each species.

The Id column identifies each record, and Species contains the actual flower names. K-Means uses the four numerical measurements for clustering, not the species labels.

4. What is K-Means clustering?

K-Means is an unsupervised learning algorithm that divides data into K clusters.

Here, K means the number of clusters we want to create.

For example, if K = 3, the algorithm tries to divide the flowers into three groups.

How does K-Means work?

1. Select the number of clusters (K).
2. Randomly initialize K centroids.
3. Calculate the distance between each data point and the centroids.
4. Assign each data point to its nearest centroid.
5. Recalculate the centroid of each cluster.
6. Repeat until the clusters stop changing significantly.

What is a centroid?

A centroid is the centre point of a cluster. It represents the average position of all data points belonging to that cluster.

5. Why do we use StandardScaler?

The four flower measurements have different ranges. StandardScaler converts them to a common scale so that a feature with larger numerical values does not dominate the distance calculation.

After scaling:

- Mean becomes approximately 0.
- Standard deviation becomes approximately 1.

Your notebook displays scaled means close to 0 and scaled standard deviations equal to 1.

6. What is the Elbow Method?

The Elbow Method is used to find the optimal number of clusters (K).

We don't always know how many clusters should be created. So, we try different values of K, such as 1, 2, 3, 4, 5, and calculate the WCSS for each value.

What is WCSS?

WCSS stands for Within-Cluster Sum of Squares. It measures how far the data points are from their own cluster's centroid.

Formula:

\[WCSS=\sum_{i=1}^{K}\sum_{x\in C_i} ||x-\mu_i||^2 \]

Where:

- \(K\) = Number of clusters.
- \(C_i\) = Data points in cluster i.
- \(x\) = A data point.
- \(\mu_i\) = Centroid of that cluster.

How to identify the elbow?

K-means Clustering: Complete Guide with Algorithm, Implementation & Best Practices - Interactive | Michael Brenndoerfer | Michael Brenndoerfer

mbrenndoerfer.com

- X-axis represents the number of clusters (K).
- Y-axis represents WCSS.
- WCSS decreases as the number of clusters increases.
- We select the point where the decrease starts slowing down significantly. This is called the elbow point.

For the Iris dataset, we generally expect around 3 clusters, because it contains three actual flower species. The elbow graph helps us decide whether K = 3 is a suitable choice.

7. What happens in your Jupyter Notebook?

Understand these practical steps in order:

1. Import libraries: NumPy, Pandas, Matplotlib, Seaborn and Scikit-learn are imported for data handling, visualization and clustering.
2. Load dataset: The notebook loads Iris.csv. If the file is unavailable, it uses the equivalent Iris dataset bundled with Scikit-learn.
3. Explore dataset: It checks the first rows, dataset shape, missing values, duplicate records, species counts and statistical summary.
4. Visualize data: It creates a correlation heatmap, box plot and pair plot to understand relationships between flower measurements.
5. Select features: It selects the four numerical flower measurements and keeps the actual species labels separately for later evaluation.
6. Scale the data: StandardScaler converts the measurements to a common scale.
7. Apply Elbow Method: It calculates WCSS for different K values and plots the elbow graph.
8. Apply K-Means: It trains the K-Means model using the selected number of clusters and assigns each flower a cluster label.
9. Evaluate clusters: It uses clustering evaluation measures and visualizations to understand the quality of the groups.

8. Important graphs and their meanings

Graph	**What it tells us**
Correlation heatmap	Relationship between flower features
Box plot	Distribution and outliers in measurements
Pair plot	Relationship between pairs of features

Elbow graph	Suitable number of clusters
Cluster scatter plot	How flowers are grouped into clusters

10. Important viva questions and answers

Q1. What is the aim of this practical?

To perform K-Means clustering on the Iris dataset and find the optimal number of clusters using the Elbow Method.

Q2. What is K-Means?

K-Means is an unsupervised machine learning algorithm used to divide data into K groups based on similarity.

Q3. What is clustering?

Clustering is the process of grouping similar data points together.

Q4. What is a centroid?

A centroid is the centre point of a cluster.

Q5. What is the Elbow Method?

It is a method used to find the optimal number of clusters by plotting WCSS against different values of K.

Q6. What is WCSS?

WCSS stands for Within-Cluster Sum of Squares. It measures the sum of squared distances between data points and their cluster centroids.

Q7. Why do we use StandardScaler?

To bring all numerical features to a common scale.

Q8. Is K-Means supervised or unsupervised?

Unsupervised learning.

Q9. How many species are present in the Iris dataset?

Three species: Setosa, Versicolor and Virginica.

Q10. What is the difference between classification and clustering?

Classification uses labelled data, whereas clustering groups data based on similarity without requiring labels.

Q11. Can K-Means work without knowing the actual species?

Yes. K-Means does not need species labels to create clusters.

Q12. What is the main disadvantage of K-Means?

We need to choose the number of clusters in advance, and the algorithm can be affected by outliers and initial centroid positions.

11. How to explain this practical to the examiner

You can speak this paragraph:

"This practical is based on K-Means clustering using the Iris dataset. The dataset contains 150 flower records with four numerical features: sepal length, sepal width, petal length and petal width. First, we load and explore the dataset. Then we perform data visualization and standardization. After that, we use the Elbow Method to identify the optimal number of clusters using WCSS. Finally, we apply the K-Means algorithm to group similar flowers and evaluate the clustering results."

12. Conclusion

Thus, we successfully perform K-Means clustering on the Iris dataset. We use the Elbow Method to estimate the optimal number of clusters and group similar flowers based on their measurements.

Most important for tomorrow's practical

Focus on these five things:

1. Meaning of clustering and K-Means.
2. Difference between supervised and unsupervised learning.
3. Meaning of centroid.
4. Elbow Method and WCSS formula.
5. Why StandardScaler is used.

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1. Euclidean Distance

Used to calculate the distance between two data points.

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

2. Centroid

The average position of all points in a cluster.

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

3. WCSS

Used in the Elbow Method to find the suitable number of clusters.

$$WCSS = \sum ||x - \mu||^2$$

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4. Silhouette Score

Measures how well a data point fits into its own cluster compared with other clusters.

$$S = \frac{b - a}{\max(a, b)}$$

Here, a is the average distance within its own cluster and b is the average distance to the nearest other cluster.

A higher silhouette score generally indicates better-separated clusters.

Prac 5

Title: Different Voting Mechanisms with AdaBoost, Gradient Tree Boosting (GBM), and XGBoost Classification on the Iris Dataset.

Let's study this practical step by step for your practical exam. I'll explain the concepts, what we do in Jupyter Notebook, the outputs and graphs, and important viva questions—not every line of code.

1. Aim of the practical

To implement and compare different ensemble learning techniques, such as AdaBoost, Gradient Boosting, XGBoost, Hard Voting, and Soft Voting, using the Iris dataset.

2. What is Ensemble Learning?

Ensemble Learning is a machine learning technique in which multiple models are combined to produce a better prediction than a single model.

In simple words: instead of depending on one model, we combine multiple models to improve prediction accuracy.

Example: If three models predict the flower type:

- Model 1 predicts Setosa.
- Model 2 predicts Versicolor.
- Model 3 predicts Setosa.

The final prediction can be Setosa because it receives the majority of votes.

3. Dataset used – Iris Dataset

The practical uses the Iris dataset, which contains information about three types of flowers.

The dataset contains measurements such as:

Feature	Meaning
Sepal Length	Length of sepal
Sepal Width	Width of sepal
Petal Length	Length of petal
Petal Width	Width of petal

Target classes:

1. Iris-setosa
2. Iris-versicolor
3. Iris-virginica

The objective is to predict the flower species using its measurements.

4. Important concepts used in this practical

A. AdaBoost

AdaBoost stands for Adaptive Boosting.

- It combines multiple weak learners to create a strong model.
- It trains models one after another.
- More attention is given to incorrectly predicted data points.
- The next model tries to correct the previous model's mistakes.

Simple example: If the first model wrongly predicts some flowers, the next model gives more importance to those wrongly predicted flowers.

B. Gradient Boosting (GBM)

GBM stands for Gradient Boosting Machine.

- It builds decision trees one after another.
- Each new tree tries to correct the errors made by previous trees.
- It combines the predictions of all trees to produce the final result.

Remember: AdaBoost focuses on wrongly classified samples, while GBM focuses on reducing prediction errors using gradients.

C. XGBoost

XGBoost stands for Extreme Gradient Boosting.

- It is an advanced implementation of Gradient Boosting.
- It builds trees sequentially.
- It uses regularization to help control overfitting.
- It is designed to provide efficient and accurate predictions.

D. Voting Classifier

A Voting Classifier combines predictions from multiple machine learning models.

There are two types:

1. Hard Voting

- Each model predicts a class.
- The class receiving the maximum number of votes becomes the final prediction.

2. Soft Voting

- Each model gives probabilities for each class.
- The probabilities are averaged.
- The class with the highest average probability becomes the final prediction.

Difference between Hard Voting and Soft Voting

Hard Voting	**Soft Voting**
Uses predicted class labels.	Uses predicted probabilities.
Selects the majority class.	Selects the class with the highest average probability.
Does not require probability estimates.	Requires probability estimates.

5. Steps performed in Jupyter Notebook

Your practical follows these main steps:

1. Import libraries: Import NumPy, Pandas, Matplotlib, Seaborn and Scikit-learn tools, including AdaBoost, Gradient Boosting, Voting Classifier and XGBoost.
2. Load the dataset: Load the Iris dataset and check its features and flower species.
3. Data preparation: Separate input features (X) and target labels (y). Encode the species labels into numerical values.
4. Visualize the data: Use a pair plot to understand relationships between the four flower measurements.
5. Split the dataset: Divide the data into training and testing sets.
6. Train models: Apply AdaBoost, Gradient Boosting and XGBoost.
7. Apply voting: Combine the models using Hard Voting and Soft Voting.
8. Evaluate models: Compare accuracy, precision, recall and F1-score. Display classification reports and confusion matrices.

The notebook uses 120 samples for training and 30 samples for testing. It also sets the random state to 42 for reproducibility.

SHRUTIKA CL1 CL2 (1)(1).pdf

6. Understanding the outputs

Your notebook produces a few important outputs.

A. Pair plot-It displays relationships between different flower measurements. Different colours represent different Iris species.

B. Confusion matrix-It compares actual flower species with predicted flower species.

- Diagonal values = correct predictions.
- Values outside the diagonal = incorrect predictions.
- Rows represent actual classes.
- Columns represent predicted classes.

C. Performance comparison graph

It compares the models using accuracy, precision, recall and F1-score. A higher score generally indicates better performance.

7. Final results of your practical

According to your notebook, the model comparison is:

Model	Accuracy
AdaBoost	96.67%
Gradient Boosting (GBM)	93.33%
XGBoost	96.67%
Hard Voting	96.67%
Soft Voting	93.33%

What does this result mean?

- AdaBoost correctly predicted 29 out of 30 test samples.
- GBM correctly predicted 28 out of 30 test samples.
- XGBoost correctly predicted 29 out of 30 test samples.
- Hard Voting correctly predicted 29 out of 30 test samples.
- Soft Voting correctly predicted 28 out of 30 test samples.

In this experiment, AdaBoost, XGBoost and Hard Voting achieved the highest accuracy of 96.67%.

9. Important viva questions and answers

Q1. What is Ensemble Learning?

It is a technique that combines multiple machine learning models to improve prediction performance.

Q2. What is AdaBoost?

AdaBoost is a boosting algorithm that gives more importance to incorrectly classified samples.

Q3. What is Gradient Boosting?

It builds models sequentially, where each new model tries to correct the errors of previous models.

Q4. What is XGBoost?

XGBoost is an advanced and efficient implementation of the Gradient Boosting algorithm.

Q5. What is Voting Classifier?

It combines predictions from multiple models to produce one final prediction.

Q6. What are the types of voting?

Hard Voting and Soft Voting.

Q7. Difference between Bagging and Boosting?

Bagging trains models independently, while Boosting trains models sequentially.

Q8. What is a weak learner?

A weak learner is a model that performs slightly better than random guessing.

Q9. Why do we split the dataset?

To train the model on one portion and test its performance on unseen data.

Q10. Which model performed best in your practical?

AdaBoost, XGBoost and Hard Voting performed best, with 96.67% accuracy.

10. How to explain this practical to the examiner

You can speak this answer during your practical exam:

"Our fifth practical is based on Ensemble Learning using the Iris dataset. First, we load and explore the dataset, which contains four flower measurements and three flower species. Then, we divide the data into training and testing sets. We apply AdaBoost, Gradient Boosting and XGBoost algorithms. After that, we combine the models using Hard Voting and Soft Voting. Finally, we evaluate their performance using accuracy, precision, recall, F1-score and confusion matrices. In our results, AdaBoost, XGBoost and Hard Voting achieved 96.67% accuracy."

8. Important formulas for viva

Accuracy

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1-score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Prac 3

Practical 6: Reinforcement Learning

CL-1 Lab Assignment 6 (Group A)

Title: Building a Tic-Tac-Toe Game Using Reinforcement Learning in Python.

This is your last practical in the CL-1 Machine Learning list. Let's understand it from the basics, in simple language.

1. Aim of the practical

To build and train an intelligent Tic-Tac-Toe agent using Reinforcement Learning and Q-Learning in Python.

2. What is Reinforcement Learning?

Reinforcement Learning (RL) is a type of machine learning in which an agent learns by interacting with an environment and receiving rewards or penalties for its actions.

In simple words, the computer learns by playing a game, making mistakes and improving its decisions.

Example: Imagine teaching a computer to play Tic-Tac-Toe:

- If it wins, it receives a positive reward.
- If it loses, it receives a negative reward.
- If the game is a draw, it receives zero reward.

By playing many games, the computer learns which moves are better.

3. Important terms in Reinforcement Learning

These five terms are very important for your viva.

Term	**Meaning**
Agent	The learner or decision-maker (computer).
Environment	The place where the agent works (Tic-Tac-Toe game).
State	The current condition of the game board.
Action	A move taken by the agent.
Reward	Feedback received after taking an action.

Example in your practical:

- Agent: X or O computer player.
- Environment: Tic-Tac-Toe board.
- State: Current arrangement of X, O and empty positions.
- Action: Selecting an empty position.
- Reward: +1 for winning, -1 for losing, 0 for a draw.

4. What is Tic-Tac-Toe?

Tic-Tac-Toe is a two-player game played on a 3 × 3 board.

- One player uses X.
- The other player uses O.
- A player wins by placing three symbols in a row, column or diagonal.
- If all positions are filled and nobody wins, the game is a draw.

Your practical creates this game in Python and trains two computer agents to learn how to play it.

5. What is Q-Learning?

Q-Learning is a Reinforcement Learning algorithm that helps an agent learn which action is best in a particular state.

It stores information in a Q-Table.

The Q-Table stores:

- Different game states.
- Available actions for each state.
- The Q-value of each action.

What is a Q-value?

A Q-value represents how good an action is in a particular state. A higher Q-value means the action is considered more promising.

For example, if the computer has three possible moves, it compares their Q-values and generally selects the move with the highest value.

6. Important Q-Learning concepts

Your practical uses three important parameters:

Parameter	Value	Meaning
Alpha (α)	0.2	Learning rate; controls how quickly the agent learns.
Gamma (γ)	0.95	Discount factor; controls the importance of future rewards.
Epsilon (ε)	1.0	Exploration rate; controls how often the agent tries random moves.

There are two important terms:

1. Exploration

The agent tries new or random actions to discover better moves.

2. Exploitation

The agent selects the best-known action based on its learned Q-values.

Initially, epsilon is 1.0, so the agent explores a lot. As training continues, epsilon decreases, and the agent increasingly uses what it has learned.

Your notebook also uses:

- Minimum epsilon = 0.05
- Epsilon decay = 0.9995

7. Q-Learning formula

This is the main formula used in Q-Learning:

Don't worry about memorizing the formula immediately. Understand that the formula updates the Q-value based on the reward and the expected future reward.

8. Steps performed in Jupyter Notebook

Your practical follows these steps:

Step 1: Import libraries

The notebook imports:

- NumPy – for numerical operations.
- Pandas – for handling data.
- Random – for selecting random moves.
- Matplotlib – for plotting graphs.
- Defaultdict – for storing Q-values.

Step 2: Create the Tic-Tac-Toe environment

A 3 × 3 game board is created using an array of nine positions.

Initially, all positions are empty, represented by zero.

- X is represented by 1.
- O is represented by -1.
- Empty position is represented by 0.

Step 3: Define game rules

The program checks:

- Available empty positions.
- Whether X has won.
- Whether O has won.
- Whether the game is a draw.
- Whether the game should continue.

Step 4: Create the Q-Learning agent

Two Q-Learning agents are created:

- Agent X
- Agent O

Both agents maintain their own Q-Tables and learn how to choose better moves.

Step 5: Train the agents

The agents play against each other repeatedly.

Your notebook trains them for 20,000 episodes.

An episode means one complete game of Tic-Tac-Toe.

During training:

- Agents select actions.
- The game environment updates the board.
- Agents receive rewards.
- Q-values are updated.
- Epsilon gradually decreases.

Step 6: Track training performance

The notebook records:

- X wins.
- O wins.
- Draws.
- Epsilon values.

This information helps us understand how training progresses.

Step 7: Test the trained agent

After training, the notebook tests the trained X agent against a random O player.

During testing, the agent uses its learned Q-values instead of exploring randomly.

9. Understanding the Jupyter Notebook outputs

Your practical produces important results and two graphs.

A. Training completed

After 20,000 games, the notebook displays:

Output	**Result**
States learned by X	2,181

States learned by O	1,907
Final epsilon	0.05

This means both agents have learned many game states, and the exploration rate has decreased to its minimum value.

B. Training performance graph

This graph shows:

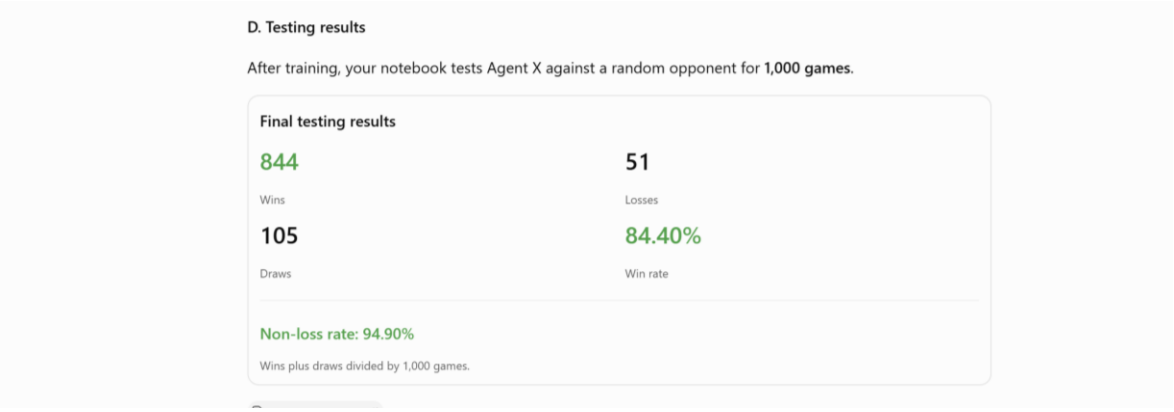
- X wins per 100 games.
- O wins per 100 games.
- Draws per 100 games.

It helps us observe the results of training over time.

C. Exploration rate decay graph

- X-axis represents training episodes.
- Y-axis represents epsilon value.
- Epsilon decreases as training progresses.
- The agent gradually shifts from exploration to exploitation.

D. Testing results



After training, your notebook tests Agent X against a random opponent for 1,000 games.

The trained agent won 844 games out of 1,000 against the random opponent. This shows that the agent learned to make better moves through training.

E. Final demonstration

Your notebook also plays one sample game between:

- Trained Agent X
- Random Agent O

In the displayed game, Agent X chooses positions 4, 5 and 6 and completes the middle row to win.

10. Important viva questions and answers

Q1. What is Reinforcement Learning?

It is a machine learning technique where an agent learns by interacting with an environment and receiving rewards or penalties.

Q2. What is Q-Learning?

Q-Learning is a model-free Reinforcement Learning algorithm that learns the best action for each state using Q-values.

Q3. What is an agent?

An agent is the learner or decision-maker that performs actions.

Q4. What is an environment?

The environment is the place where the agent performs actions. In our practical, it is the Tic-Tac-Toe game.

Q5. What is a reward?

A reward is feedback given to the agent after an action.

Q6. What is a Q-Table?

A Q-Table stores Q-values for different states and actions.

Q7. What is epsilon?

Epsilon controls exploration, meaning how often the agent chooses random actions.

Q8. What is alpha?

Alpha is the learning rate. It controls how much new information changes the Q-value.

Q9. What is gamma?

Gamma is the discount factor. It determines the importance of future rewards.

Q10. What is an episode?

One complete game or interaction sequence is called an episode.

Q11. What is self-play?

Self-play means two learning agents play against each other to improve their strategies.

Q12. How many episodes are used in your practical?

20,000 training episodes.

Q13. How many games are used for testing?

1,000 games against a random opponent.

Q14. What is the final win rate?

84.40%.

Q15. What is the difference between exploration and exploitation?

Exploration tries new actions, whereas exploitation selects the best-known action.

11. How to explain this practical to the examiner

You can speak this answer during your practical exam:

"Our sixth practical is based on Reinforcement Learning using a Tic-Tac-Toe game. First, we create a 3 × 3 game environment and define the rules. Then, we create two Q-Learning agents, X and O, with separate Q-Tables. The agents learn by playing against each other for 20,000 episodes. They receive rewards for winning and penalties for losing, and update their Q-values using the Q-Learning algorithm. After training, we test Agent X against a random opponent for 1,000 games. Our trained agent wins 844 games, achieving an 84.40% win rate."

12. Quick revision before your exam

Topic	**Remember this**
Algorithm	Q-Learning
Learning type	Reinforcement Learning
Game	Tic-Tac-Toe
Board	3 × 3
Agents	X and O
Alpha	0.2
Gamma	0.95
Initial epsilon	1.0
Minimum epsilon	0.05
Training	20,000 episodes
Testing	1,000 games
Win rate	84.40%